

Adversarial Robustness

UMC 203 Project Presentation

Shubham Harikrishnabhai Meghani Chimirela Karthikeshwar
K Kausigan Sudarshana Chaitanya B N Saifil Khan Pathan

IISc Bangalore

Introduction

Adversarial Robustness

- Adversarial robustness is the ability of a machine learning model to make correct predictions even when the input is slightly and intentionally perturbed.
- These perturbations are often small enough to be almost invisible to humans, but they can still cause a model to misclassify.
- This makes robustness an important requirement for reliable AI systems, especially in security-related settings.
- In this project, we study adversarial robustness from multiple complementary perspectives: understanding why adversarial transfer is asymmetric, and how to design more robust models

1. **Understanding Vulnerability:** Why adversarial examples transfer asymmetrically across models?
2. **Improving Robustness:** Using diffusion models and incremental adversarial training

Goal

Understand the mechanisms behind adversarial vulnerability and design stronger, more reliable defense strategies.

Understanding and Improving Robustness

Asymmetric Adversarial Transferability: Key Definitions

Adversarial Example: Given a model f and input (x, y) ,

$$x^{adv} = x + \delta, \quad \|\delta\|_{\infty} \leq \epsilon$$

such that

$$f(x^{adv}) \neq y$$

Transferability: Adversarial examples crafted on a source model f_s also misclassify target model f_t :

$$f_t(x_{f_s}^{adv}) \neq y$$

Fooling Rate (Transfer Success Rate):

$$\text{FR}(f_s \rightarrow f_t) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(f_t(x_i^{adv}) \neq y_i)$$

Asymmetric Adversarial Transferability: Key Definitions

Asymmetry in Transferability:

$$\text{FR}(f_s \rightarrow f_t) \neq \text{FR}(f_t \rightarrow f_s)$$

$$\Delta(f_s, f_t) = \text{FR}(f_s \rightarrow f_t) - \text{FR}(f_t \rightarrow f_s)$$

Gradient Cosine Similarity:

$$\cos(\theta) = \frac{\nabla_x L(f_s(x), y) \cdot \nabla_x L(f_t(x), y)}{\|\nabla_x L(f_s(x), y)\|_2 \cdot \|\nabla_x L(f_t(x), y)\|_2}$$

Measures alignment between loss gradients of two models. Higher similarity suggests perturbations may transfer more effectively.

Gradient Norm:

$$\|\nabla_x L(f(x), y)\|_2 = \sqrt{\sum_i \left(\frac{\partial L}{\partial x_i}\right)^2}$$

Model Selection for Adversarial Behavior Analysis

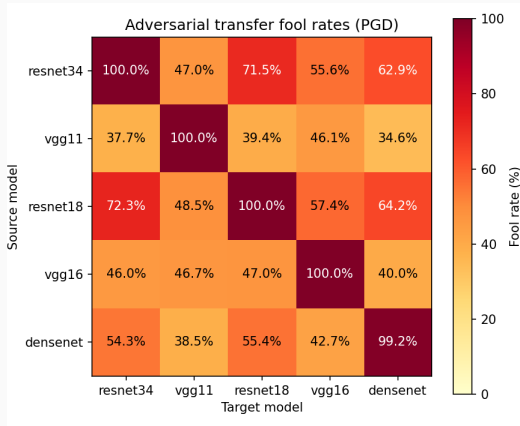
Goal: Isolate how architecture and capacity affect transfer asymmetry

Model	Architecture	# Parameters
ResNet18	Residual CNN	11.7M
ResNet34	Residual CNN	21.8M
VGG11	Plain CNN	132M
VGG16	Plain CNN	138M
DenseNet121	Dense Connectivity	8.0M

Why these models?

- Wide variation in **model capacity** (8M $\rightarrow$ 138M parameters)
- Different **connectivity patterns** (residual vs dense vs plain)

Adversarial Transferability (PGD Results)



Key Findings

- **Asymmetry exists:**
 $f_s \rightarrow f_t \neq f_t \rightarrow f_s$
- **Large effect:** up to 10–15% difference
- **Structured behavior:**
not random $\Rightarrow$ architecture-driven
- **Direction matters:**
transfer is not bidirectional

Effect of Architecture on Transfer Asymmetry

Key Observations (from transfer matrix):

- **ResNets (Residual CNNs):**
 - Often generate perturbations that transfer well to other models
 - But transfer is not symmetric in reverse direction
- **VGG (Plain CNNs):**
 - Show moderate transfer, but weaker in some directions (e.g., to ResNets)
 - e.g., **ResNet $\rightarrow$ VGG $>$ VGG $\rightarrow$ ResNet**
- **DenseNet:**
 - Strong variability across pairs (inconsistent $A \rightarrow B$ vs $B \rightarrow A$)
 - Complex connectivity leads to unstable transfer patterns

Takeaway:

- Transferability depends on **interaction between architectures**
- Suggests an underlying factor (beyond architecture alone) drives asymmetry

Why Gradient Norm Controls Attack Strength

Setup: Input a , loss $\mathcal{L}(f(a), y)$

First-order approximation:

$$\mathcal{L}(a + \delta) \approx \mathcal{L}(a) + \delta^\top \nabla_a \mathcal{L}(a)$$

FGSM attack:

$$\delta = \epsilon \cdot \text{sign}(\nabla_a \mathcal{L}(a))$$

Substitute:

$$\mathcal{L}(a + \delta) \approx \mathcal{L}(a) + \epsilon \cdot \text{sign}(\nabla_a \mathcal{L}(a))^\top \nabla_a \mathcal{L}(a) = \mathcal{L}(a) + \epsilon \|\nabla_a \mathcal{L}(a)\|_1$$

Norm bound:

$$\|\nabla_a \mathcal{L}(a)\|_1 \geq \|\nabla_a \mathcal{L}(a)\|_2 \Rightarrow \mathcal{L}(a + \delta) \geq \mathcal{L}(a) + \epsilon \|\nabla_a \mathcal{L}(a)\|_2$$

Key Insight:

$$\text{Loss increase} \geq \epsilon \|\nabla_a \mathcal{L}(a)\|_2$$

Transfer Depends on Gradient Scaling

Two models: f and g

- Assume: $f(a) \approx g(a)$ on clean input

Assume gradient norm relation:

$$\|\nabla_a \mathcal{L}_f(a)\|_2 = \alpha \cdot \|\nabla_a \mathcal{L}_g(a)\|_2$$

Attack crafted on f :

$$\delta = \epsilon \cdot \text{sign}(\nabla_a \mathcal{L}_f(a))$$

Effect on g :

$$\mathcal{L}_g(a + \delta) \leq \mathcal{L}_g(a) + \epsilon \|\nabla_a \mathcal{L}_g(a)\|_2$$

Substitute:

$$\mathcal{L}_g(a + \delta) \approx \mathcal{L}_g(a) + \epsilon \cdot \text{sign}(\nabla_a \mathcal{L}_f(a))^T \nabla_a \mathcal{L}_g(a)$$

Gradient Norm Mismatch Causes Asymmetry

Key Observation:

- Transfer strength scales with gradient norm ratio α

If $\alpha \neq 1$:

- Attack from $f \rightarrow g$ is scaled by α
- Reverse attack $g \rightarrow f$ scaled by $\frac{1}{\alpha}$

$$\Rightarrow \text{FR}(f \rightarrow g) \neq \text{FR}(g \rightarrow f)$$

Conclusion:

- Gradient norm mismatch $\Rightarrow$ unequal perturbation strength
- $\Rightarrow$ **directional asymmetry in transferability**

Empirical Evidence:

$$r = 0.96, \quad \rho = 0.88$$

Role of Gradient Direction in Transfer

Question:

- Does gradient direction affect transfer asymmetry?

Cosine Similarity:

$$\cos(\theta) = \frac{\nabla_x \mathcal{L}(f_s(x), y) \cdot \nabla_x \mathcal{L}(f_t(x), y)}{\|\nabla_x \mathcal{L}(f_s(x), y)\| \|\nabla_x \mathcal{L}(f_t(x), y)\|}$$

Meaning:

- High $\cos(\theta) \rightarrow$ aligned gradients
- Low $\cos(\theta) \rightarrow$ misaligned gradients

Key Insight:

- Cosine similarity is **symmetric**

$$\cos(f_s, f_t) = \cos(f_t, f_s)$$

- Cannot explain **directional asymmetry**

Does Gradient Direction Explain Asymmetry?

Experiment:

- Compute cosine similarity between model gradients
- Compare with observed transfer asymmetry

Result:

$$r \approx -0.10, \quad \rho \approx 0.10$$

- No meaningful correlation

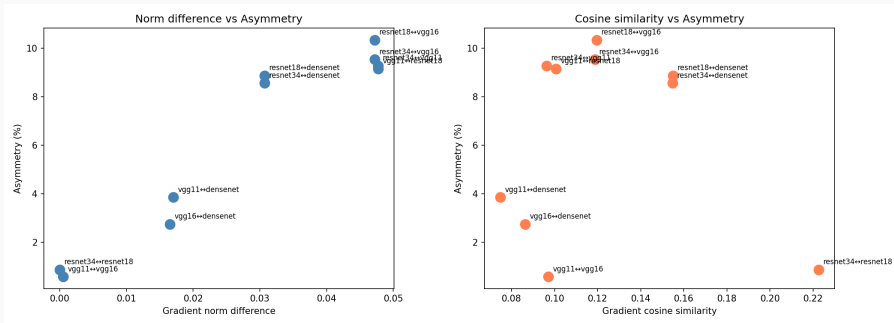
Interpretation:

- Even with aligned gradients, transfer remains asymmetric
- Direction does not control transfer strength

Conclusion:

- Gradient direction is not the driver of asymmetry

Gradient Properties vs Asymmetry



Correlation Analysis:

Metric	Pearson (r)	Spearman (ρ)	Strength
Norm diff vs Asym	0.96	0.88	Strong
Cosine sim vs Asym	-0.10	0.10	None
Param ratio vs Asym	0.25	0.31	Weak

Improving Robustness - Using Diffusion Models

Core Idea: Better synthetic data $\Rightarrow$ better robustness

- **More Data:** Diffusion models generate large, labeled datasets for AT
- **Better Quality:** Lower FID $\Rightarrow$ data closer to real distribution
- **Less Overfitting:** Reduces robust overfitting, improves generalization
- **Stronger Boundaries:** Adds hard examples $\Rightarrow$ smoother decision boundaries

Based on *Better Diffusion Models Further Improve Adversarial Training*

Improving Robustness - Using IncAT

Goal: Improve robustness *without retraining the whole model*

- **Step 1: Pretrain Model** Train the model on clean data
- **Step 2: Estimate Importance** Compute Fisher Information $\Rightarrow$ identify important parameters
- **Step 3: Generate Adversarial Data** Use FGSM / PGD / BIM to create adversarial samples
- **Step 4: Incremental Update** Train only with adversarial data + constrain important weights
- **Key Idea:** Preserve old knowledge while learning adversarial robustness

Loss Function:

$$L(\theta) = L_B(\theta) + \sum_i \frac{\lambda}{2} F_i (\theta_i - \theta_{A,i}^*)^2$$

Based on *An incremental adversarial training method enables timeliness and rapid new knowledge acquisition*

Experimental Setup

- **Data & Model:** CIFAR-10, ResNet-18, 30% & 70% synthetic data
- **Baseline:** TRADES ($\beta = 5$), 50 and 100 epochs
- **Attack:** PGD (ℓ_∞ , $\epsilon = 8/255$, step 2/255)
- **Optimization:** SGD + Nesterov, lr=0.02, cosine schedule
- **IncAT:** Adversarial fine-tuning + Fisher regularization
- **Evaluation:** AutoAttack (clean & robust accuracy)

Results: Diffusion + IncAT

Experiment	Syn %	Ep	PGD	ϵ	λ	Clean	Robust Acc
TRADES (A)	30	50	2	0.031	-	90.17	39.32
INCAT on A	30	+50	10	0.031	10^{-5}	85.95	46.06
TRADES (B)	70	100	8	0.031	-	85.16	52.26
INCAT on B	70	+50	10	0.031	10^{-5}	80.43	47.44

Observations

After training with TRADES on mixed (real + synthetic) data, we observe that the weaker model improves after incremental adversarial training (IncAT), whereas the already robust model degrades under the same process.

Effect of TRADES:

- TRADES enforces consistency between predictions on clean and adversarial inputs.
- This results in a smoother decision boundary around the data which increases the margin around samples thereby improving robustness

Effect of Fisher Regularization (INCAT):

- Fisher regularization constrains updates to parameters that are important for the current model.
- Adversarial training, however, requires modifying these parameters to improve robustness.
- This creates a conflict between stability and adaptability.
- Consequently, the model cannot effectively refine the decision boundary and the robustness degrades.

Key Takeaways

- We analyzed how different factors such as gradient norm and gradient cosine similarity affect robustness and transferability.
- Our experiments also highlight the critical role of the decision boundary in determining adversarial robustness.

Thank You !

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